1 乘法原理

乘法原理用于计算 Descartes 积的势. 为了区分 Descartes 积, 这里用 $\bigotimes$ 表示对基数的求积.

定理 1.1

任给集合 A, B, 有 $|A \times B| = |A| \cdot |B|$.

证明.

记 A, B 的势分别为 κ, λ, 则显然 $A \times B \approx \kappa \times \lambda \approx \kappa\lambda$. $\square$

定理 1.2

任给有限集 X 和 X 上的映射 f, 有 $\left|\prod_{x \in X} f(x)\right| = \bigotimes_{x \in X} |f(x)|$.

证明.

不妨设 X 非空, 取双射 $\varphi : |X| \to X$. 对 $n < |X|$ 归纳证明 $\left|\prod_{i=0}^{n} f\varphi(i)\right| = \bigotimes_{i=0}^{n} |f\varphi(i)|$.

首先, $n = 0$ 时

$$\left|\prod_{i=0}^{0} f\varphi(i)\right| = \left|{}^{1} f\varphi(0)\right| = |f\varphi(0)| = \bigotimes_{i=0}^{0} |f\varphi(i)|\,;$$

其次, 假设命题对 n 成立且 $n + 1 < |X|$, 那么易证

$$\prod_{i=0}^{n+1} f\varphi(i) \to \prod_{i=0}^{n} f\varphi(i) \times f\varphi(n+1), \quad p \mapsto (p\restriction_{n+1}, p(n+1))$$

是双射, 从而

$$\left|\prod_{i=0}^{n+1} f\varphi(i)\right| = \left|\prod_{i=0}^{n} f\varphi(i) \times f\varphi(n+1)\right| = \bigotimes_{i=0}^{n} |f\varphi(i)| \cdot |f\varphi(n+1)| = \bigotimes_{i=0}^{n+1} |f\varphi(i)|,$$

归纳成立, 取 $n = |X| - 1$ 即得 $\left|\prod_{x \in X} f(x)\right| = \left|\prod_{i=0}^{|X|-1} f\varphi(i)\right| = \bigotimes_{i=0}^{|X|-1} |f\varphi(i)| = \bigotimes_{x \in X} |f(x)|$. $\square$